

- 11 Two satellites X and Y are orbiting at heights of $5R$ and $9R$ above the surface of the Earth respectively, where R is the radius of the Earth.

What is the ratio of their orbital speeds v_X/v_Y ?

- A** 1.29 **B** 1.34 **C** 1.67 **D** 1.80

$$G M m / r^2 = m v^2 / r \rightarrow G M / r = v^2$$

$$\rightarrow v_X / v_Y = (r_Y / r_X)^{1/2} = (10R / 6R)^{1/2} = 1.29$$

- 12 The gravitational potentials on the surface of a planet P and on the surface of its moon Q are